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17-09-26 agile lab

PROJECT 1: Parameterized Build Pipeline

Jenkins-pipeline-lab3 / Jenkinsfile

krithikashree1957 Rename jen to Jenkinsfile 68249db · 3 minutes ago History

Code Blame 32 lines (28 loc) · 763 Bytes

```
1 pipeline {
2   agent any
3
4   parameters {
5     choice(
6       name: 'ENVIRONMENT',
7       choices: ['dev', 'staging', 'prod'],
8       description: 'Select the deployment environment'
9     )
10  }
11
12  stages {
13    stage('Checkout') {
14      steps {
15        git branch: 'main',
16          url: 'https://github.com/krithikashree1957/Jenkins-pipeline-lab3.git'
17      }
18    }
19
20    stage('Show Parameter') {
21      steps {
22        echo "Selected environment: ${params.ENVIRONMENT}"
23      }
24    }
25
26    stage('Build for Environment') {
27      steps {
28        echo "Building the application for the ${params.ENVIRONMENT} environment..."
29      }
30    }
31  }
32 }
```

Jenkins / project1

Status

Changes

Build with Parameters

Configure

Delete Pipeline

Full Stage View

Favorite

Job Config History

Open Blue Ocean

Stages

Rename

Pipeline Syntax

Pipeline project1

This build requires parameters:

ENVIRONMENT
Select the deployment environment

dev

Build

Parameter: dev

```
Selected environment: dev
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build for Environment)
[Pipeline] echo
Building the application for the dev environment...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Parameter: Staging

```
Selected environment: staging
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build for Environment)
[Pipeline] echo
Building the application for the staging environment...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Parameter: prod

```
Selected environment: prod
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build for Environment)
[Pipeline] echo
Building the application for the prod environment...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

PROJECT 2: Archive Build Artifacts Pipeline

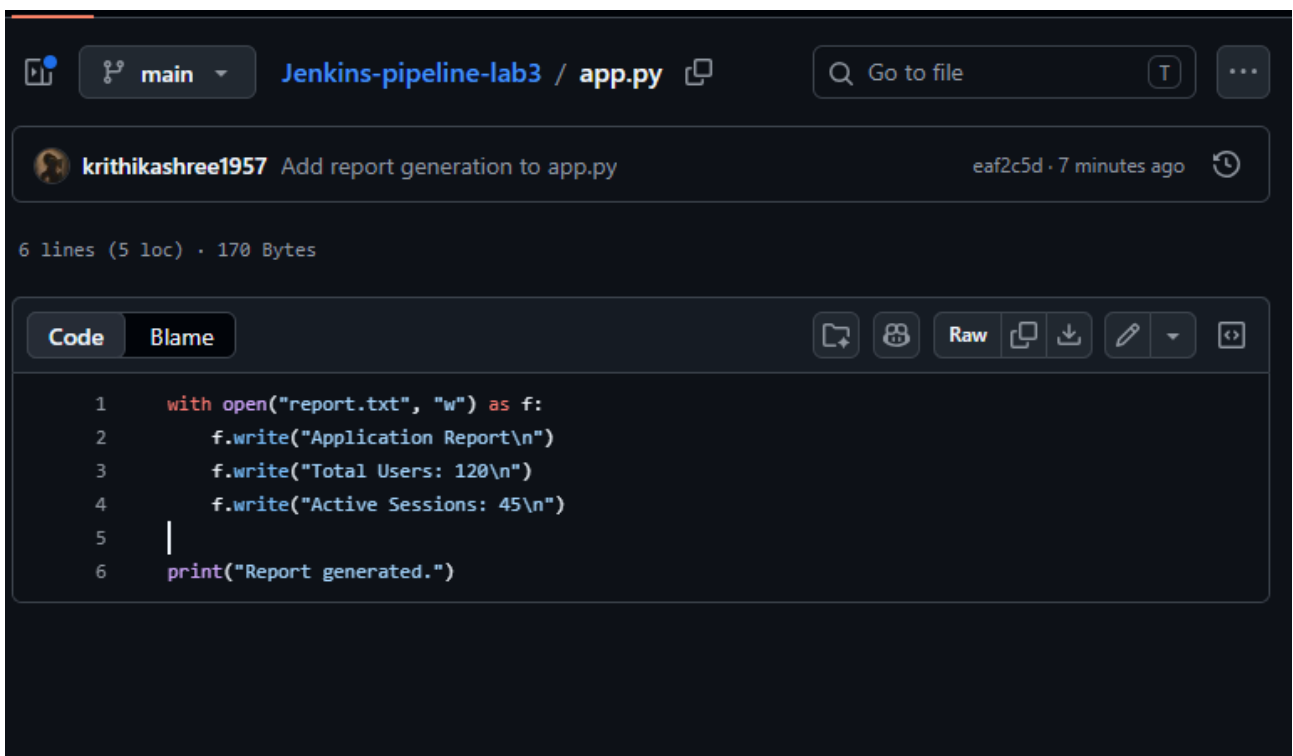
pipeline script:

```
pipeline {
  agent any

  stages {
    stage('Checkout') {
      steps {
        checkout scmGit(branches: [[name: '**']], extensions: [], userRemoteConfigs: [[url:
'https://github.com/krithikashree1957/Jenkins-pipeline-lab3.git']])
      }
    }

    stage('Generate Report') {
      steps {
        bat 'python app.py'
      }
    }

    stage('Archive Report') {
      steps {
        archiveArtifacts artifacts: 'report.txt', followSymlinks: false
      }
    }
  }
}
```



Archive Report:

Jenkins / project2 / #1

Status

- Changes
- Console Output
- Edit Build Information
- Delete build '#1'
- Polling Log
- Timings
- Git Build Data
- Rebuild
- Open Blue Ocean
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces

✓ #1 (Sep 17, 2026, 10:47:59 AM)

Build Artifacts
report.txt 59 B view

Started by an SCM change
Started by user Krithika shree K

This run spent:

- 0.29 sec waiting;
- 2.5 sec build duration;
- 2.8 sec total from scheduled to completion.

Revision: eaf2c5de10709e57faee2a05cca2d658f947952d
Repository: <https://github.com/krithikashree1957/Jenkins-pipeline-lab3.git>

- origin/main

No changes.

report.txt content:

```
Application Report
Total Users: 120
Active Sessions: 45
```

after changing total user to 200

```
Application Report
Total Users: 200
Active Sessions: 45
```

PROJECT 3: Parallel Stages Pipeline

pipeline script:

```
pipeline {
  agent any

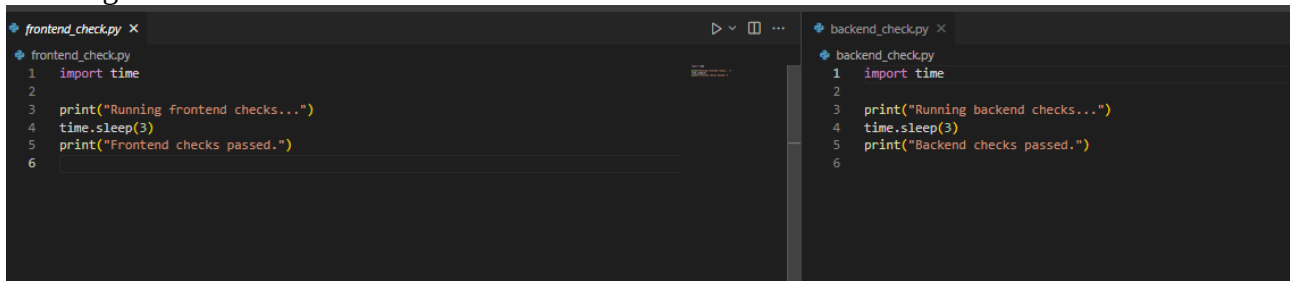
  stages {
    stage('Checkout') {
      steps {
        git branch: 'main',
            url: 'https://github.com/krithikashree1957/Jenkins-pipeline-lab3.git'
      }
    }

    stage('Parallel Checks') {
      parallel {
        stage('Frontend Check') {
          steps {
            bat 'python frontend_check.py'
          }
        }

        stage('Backend Check') {
          steps {
            bat 'python backend_check.py'
          }
        }
      }
    }

    stage('Summary') {
      steps {
        echo 'Both frontend and backend checks are complete.'
      }
    }
  }
}
```

files in github:



```
frontend_check.py
1 import time
2
3 print("Running frontend checks...")
4 time.sleep(3)
5 print("Frontend checks passed.")
6

backend_check.py
1 import time
2
3 print("Running backend checks...")
4 time.sleep(3)
5 print("Backend checks passed.")
6
```

console output:

```
[Pipeline] bat
[Frontend Check]
C:\ProgramData\Jenkins\.jenkins\workspace\project3>python frontend_check.py
[Backend Check]
C:\ProgramData\Jenkins\.jenkins\workspace\project3>python backend_check.py
[Frontend Check] Running frontend checks...
Frontend checks passed.
[Backend Check] Running backend checks...
Backend checks passed.
[Pipeline] }
[Pipeline] }
[Pipeline] // stage
[Pipeline] // stage
[Pipeline] }
[Pipeline] }
[Pipeline] // parallel
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Summary)
[Pipeline] echo
Both frontend and backend checks are complete.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

PROJECT 4: Conditional Stage Execution Pipeline

pipeline script:

```
pipeline {
  agent any

  parameters {
    booleanParam(
      name: 'RUN_EXTRA_CHECK',
      defaultValue: true,
      description: 'Run the extra check stage'
    )
  }

  stages {
    stage('Checkout') {
      steps {
        checkout scmGit(branches: [[name: '**']], extensions: [], userRemoteConfigs: [[url:
'https://github.com/krihikashree1957/Jenkins-pipeline-lab3.git']])
      }
    }

    stage('Build') {
      steps {
        bat 'python -m py_compile app_project4.py'
```

```

        echo 'Build successful: app_project4.py compiled with no syntax errors'
    }
}

stage('Extra Check') {
    when {
        expression {
            params.RUN_EXTRA_CHECK == true
        }
    }

    steps {
        echo 'Running extra check: verifying greet() output format...'
        bat 'python -c "from app_project4 import greet; print(greet(\'Student\'))"'
    }
}
}
}

```

initial build:

```

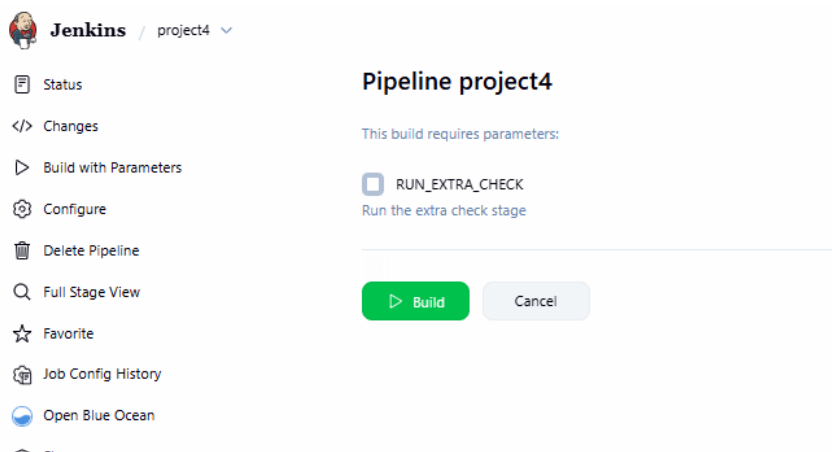
Commit message: "Add greet function to return a greeting message"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] bat

C:\ProgramData\Jenkins\jenkins\workspace\project4>python -m py_compile app_project4.py
[Pipeline] echo
Build successful: app_project4.py compiled with no syntax errors
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Extra Check)
[Pipeline] echo
Running extra check: verifying greet() output format...
[Pipeline] bat

C:\ProgramData\Jenkins\jenkins\workspace\project4>python -c "from app_project4 import greet; print(greet('Student'))"
Hello, Student
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```

build with parameter:(unchecking RUN_EXTRA_CHECK)



Jenkins / project4

- Status
- Changes
- Build with Parameters
- Configure
- Delete Pipeline
- Full Stage View
- Favorite
- Job Config History
- Open Blue Ocean

Pipeline project4

This build requires parameters:

- ☐ RUN_EXTRA_CHECK
Run the extra check stage

Build Cancel

EXTRA CHECK SKIPPED:

```
C:\ProgramData\Jenkins\.jenkins\workspace\project4>python -m py_compile app_project4.py
[Pipeline] echo
Build successful: app_project4.py compiled with no syntax errors
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Extra Check)
Stage "Extra Check" skipped due to when conditional
[Pipeline] getContext
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

PROJECT 5: Custom Environment Variables Pipeline

pipeline script:

```
pipeline {
    agent any

    environment {
        APP_NAME = 'GradeBookApp'
        APP_VERSION = '1.0.0'
    }

    stages {
        stage('Checkout') {
            steps {
                checkout scmGit(branches: [[name: '**']], extensions: [], userRemoteConfigs: [[url:
'https://github.com/kritihikashree1957/Jenkins-pipeline-lab3.git']])
            }
        }

        stage('Show App Info') {
            steps {
                echo "Building ${env.APP_NAME}, version ${env.APP_VERSION}"
            }
        }

        stage('Build') {
            steps {
                bat 'python -m py_compile app_project5.py'
                echo "${env.APP_NAME} version ${env.APP_VERSION} compiled successfully."
            }
        }
    }
}
```

```
app_project5.py X
app_project5.py
1 def add(a, b):
2     return a + b
3
```

Console output:

```
C:\ProgramData\Jenkins\.jenkins\workspace\project5>python -m py_compile app_project5.py
[Pipeline] echo
GradeBookApp version 1.0.0 compiled successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

changing APP_VERSION = '2.0.0' in pipeline script:

```
pipeline {
    agent any

    environment {
        APP_NAME = 'GradeBookApp'
        APP_VERSION = '2.0.0'
    }

    stages {
```

```
C:\ProgramData\Jenkins\.jenkins\workspace\project5>python -m py_compile app_project5.py
[Pipeline] echo
GradeBookApp version 2.0.0 compiled successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```